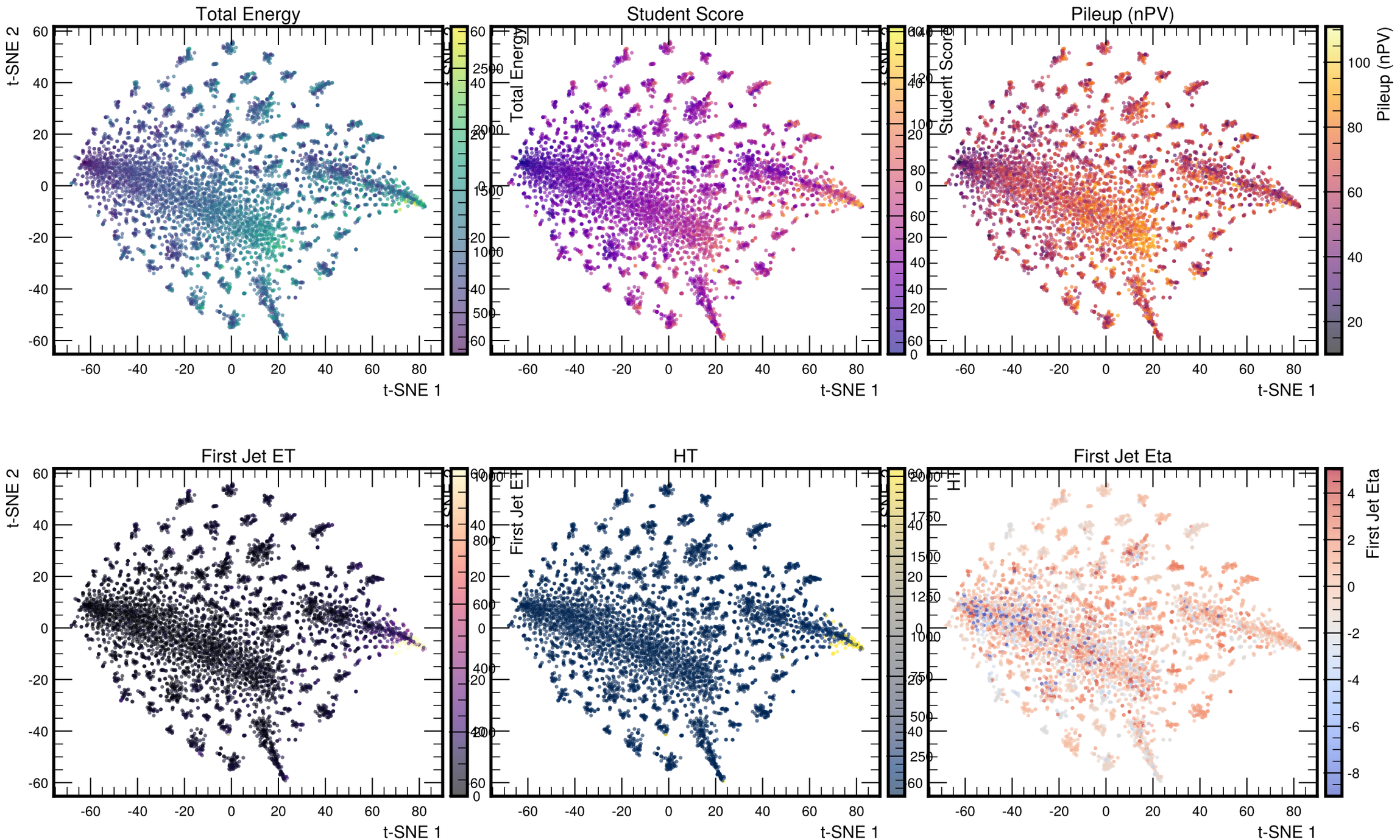hto2longlivedto4b